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Team Members: Myself
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Demonstration Method: Through a video per professor permission.

1. Design

1.1 Hardware Design

I don’t think there is anything to say about the hardware design. It is just wiring one component and even the number is provided. There is no room to get creative here.

There is a slight problem with the hardware: my kit does not contain MPU-6050 but MPU-9250 instead. MPU-9250 is more modern with more features. The pins are still the same so everything should work just fine… NOT… MPU-9250 self-reports as a different model (as expected) and the Arduino imu_stream code wants me to have MPU-6050… I will talk more on it in section 2. My journal also contains my state of mind at that moment.

1.2 Software Design

Instructions on how to setup the project is very detailed and sufficient. All of the work is to fill in the blank. And funny enough, all the answers are already given. I only need to translate natural language into programming language. And most of the time, I don’t even have to do that because the natural language instruction happens to include the programming language solution. I literally finished the entire programming part in 2 hours. No thinking, no searching, just reading the instructions that happen to spoil the final code.

There are two parts to the lab: Arduino code and Python code. Arduino code is responsible for *translating* sensor data and serialize it to Python code. Python code is responsible for *interpreting* sensor data. Arduino code is very basic and Python code is more complicated to make sense of the data using trigonometry and graph and whatnot.

1.3 Result

All the steps are done and I can understand the sensor data. Not just that, for each according steps, things just look right. For example, the filter step makes the data look smooth and “right” and it looks right to me.

The ultimatum to the project is Marble Madness… Oh wait, I just read the code for it while I am writing this, it actually uses nothing from previous step and just implements its own filter and activity detection?? I had thought that it would use previous step implementations for the game itself… It does not. Well, there is no ultimatum to the project, I guess. I did all the work and they look all right to me.

For the activity classifier, I just play around with the number. But it was not enough to I decided to change the implementation a bit. SHAKING is valid if there is one tick with high magnitude in the window rather than everything in the window is high magnitude. There is not much reasoning behind it beside I like that more than the other implementation. The magnitude number is tweaked a lot. In general, the numbers are more sensitive than the default number, meaning number for SHAKING is a lot lower. Again, the reason behind my numbers is purely empirical.

Back to Marble Madness, I tried it again during the demonstration video and the drifting issue is very bad. But that is an easy fix with lower alpha value.

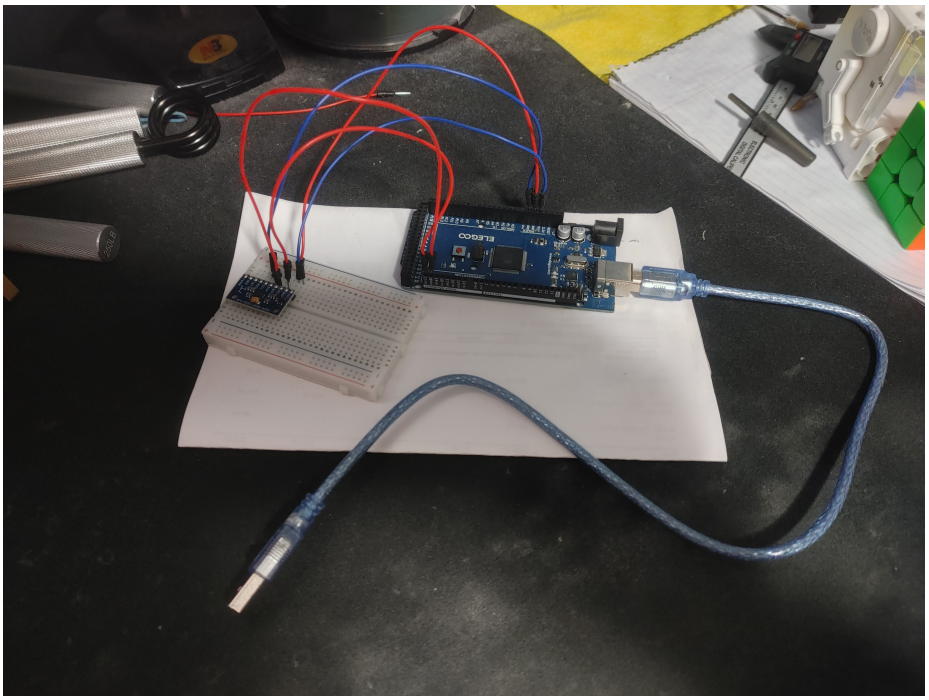
2. Problems Encountered and Solved

As mentioned, I don’t have MPU-6050 but MPU-9250 instead. In the Arduino imu_stream code, it polls for WHO_AM_I register. For the longest time, I could not figure out what the error is because I don’t have Arduino serial monitor on to check the error. Took me a whole day to decide enabling that and I see the error right away.

No problems encountered since then because this lab is very simple and straightforward.

3. Personal Contribution to the Lab

I did all the work by myself.



4. Lessons Learned

It is neat to know how a sensor reads data. I have been exposed to something similar to this before but it is more related to data sent over the network. Back in the day when internet was slow, online video games cannot afford to send a whole 32 bit for a float number. It turns out, what developers back then did is to use single point (decimal) number. It has a bit count and divisor. This is not so much different. It is a single point number.

I also never integrate or calculate derivative with code before. I know how they works, I know what integration does (definite integral is just the sum under the curve). It just never strikes to me that concept could be directly translated to code like that. I have dealt (“dealt” is an anagram of “delta”) with delta before and they are about video game timing system. Never occurred to me that delta is just a discrete way to calculate integration.